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PI Name:	Fornace, Albert M.D.		
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Comments:	http://www9.georgetown.edu/		
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Key Personnel Changes/Previous PI:	February 2021 report: Kamal Datta moved to NIH; Shubhankar Suman added as Co-Investigator in the year-6 of this NASA Specialized Center of Research (NSCOR).		
COI Name (Institution):	Brenner, David Ph.D. (Columbia University) Meltzer, Paul M.D., Ph.D. (National Cancer Institute, NIH) Shay, Jerry Ph.D. (University of Texas Southwestern Medical Center at Dallas) Suman, Shubhankar Ph.D. (Georgetown University)		
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Task Description:	Considering the high spontaneous incidence of gastrointestinal (GI) and lung cancer, and even a modest increase by space radiation exposure could have a significant effect on astronauts' health risk estimates during and after long-duration manned space flights. However, there is substantial uncertainty for GI and lung cancer risk estimation from space radiation due to the lack of in vivo human data. Our overall goal is to specifically investigate cellular and molecular hypothesis-driven mechanisms in relevant mouse models that will contribute to carcinogenic risk estimates of GI and lung cancer after exposure to space radiation beams prioritized by NASA's Human Research Program (HRP). Specifically, we seek to determine incidence, tumor frequency, and grade as well as identify molecular perturbations in the target cells associated GI and lung tumorigenesis through monitoring of changes in the tumor number, histology, and gene/protein expression of the proposed model system after exposure to space-radiation type beams. The overall hypothesis is that neutrons and mixed beam galactic cosmic radiation (GCR) type radiation are more carcinogenic due to important qualitative differences from gamma rays where human exposure data are available. Molecular events triggered by space radiation will be investigated in-depth and the mechanistic basis for persistent aberrant cell-signaling after space radiation will be dissected. Both quantitative and qualitative murine tumor data will contribute to the modeling of human risks for GI and lung cancer by space travel. We plan to test our hypothesis by pursuing the following specific aims: Aim 1. Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2. Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation. Aim 3. Quantitatively assess lung tumorigenesis and test potential countermeasures in lung cancer susceptible mouse model. Aim 4. Development of mathematical modeling for cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S., and the American Cancer Society estimates that there will be 147,950 new cases of colorectal cancer (CRC) with 53,200 persons predicted to die of the disease in 2020. Overall, the lifetime risk of developing colorectal cancer is about 1 in 23 (4.4%) for men and 1 in 25 (4.1%) for women. Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future crewed space missions. In addition to risk estimation, studies on the persistence of oxidative and inflammatory stress after HZE (high energy particle) radiation and its role in driving gastrointestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and carcinogenesis. The significance and deliverable of this project are to improve the estimates of GI cancer risk in astronauts and to identify and test plausible targets for the development of mitigation strategies.
	Task Progress: Pointwise progress made for each aim/project is summarized below: GI-tumorigenesis study (Project-1, Georgetown University): In the last six years of NSCOR funding, we have successfully applied mouse model approaches to acquire quantitative and qualitative GI tumorigenesis data to address prevailing uncertainties in risk modeling. Our overall goal is to determine the relative biological effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation. The risk estimates can then be extrapolated to space radiation using a RBE scaling factor. Our strategy is to i] quantitatively analyze tumor incidence and grade following space type radiation exposure. ii] Assess potential dose rate and gender variation. iii] Delineate cancer progression-specific biological signatures associated with space radiation exposure. Since low-linear energy transfer (LET) gamma radiation risk estimates are available for human populations from studies in atom bomb cohorts and like, we will then be able to combine our data to model the relative risk for space radiation. Pointwise details of progress made for GI tumorigenesis study are provided below: 1. Beam run participation in 2020: Georgetown University team participated in two-beam runs (NASA Space Radiation Laboratory (NSRL) 20B and 20C)), while planned experiments in the Spring-run (20A) were canceled due to COVID-19 restrictions at Brookhaven National Laboratory (BNL)/NSRL. 2. Dose-rate effect analysis using acute and chronic full GCR sim: In order to understand dose rate effects on GI-tumorigenesis after full-spectrum GCR sim exposures, we completed our initial study using 50 cGy dose of acute and chronic full-spectrum (33-ion) GCR beams during the NSRL 20B run. This initial study using a limited number of mice revealed a good signal after both acute and chronic full-spectrum GCR sim exposures, relative to controls. However, additional data from more mice are required; this initial study provided crucial insight and validated our mouse model for future experiments using lower doses of chronic full spectrum GCR beams. 3. Role of HZE in GCR-induced GI-tumorigenesis: Our previous study comparing quantitative tumor data obtained from the equivalent doses of a single HZE beam to the 4-ion mixed-field GCR-induced tumorigenesis revealed an association between higher intestinal tumor number and dose received from the HZE fraction of mixed-field GCR. Therefore, in order to understand the relative contributions of HZEs components in a full GCRsim induced tumorigenesis, we also exposed male Apc1638N/+ to p+He (45.7 cGy) and HZE only (4.3 cGy) during NSRL 20C run and quantitative tumorigenesis data will be obtained in early 2021. Once quantitative data from p+He (45.7 cGy) and HZE only (4.3 cGy) exposures are available, a comparison with 50 cGy full-GCRsim is planned to understand and model the relative role of HZE in GCR-induced tumorigenesis. 4. Role of non-targeted effects (NTEs) in HZE-induced tumorigenesis: In order to understand the role of non-targeted effects (NTEs) in HZE-induced tumorigenesis, male Apc1638N/+ mice (6-8 weeks) were whole-body exposed to sham-radiation, gamma-rays, and 28Si-ion at 10 cGy dose and mice were sacrificed at 60 d and 150 d after radiation exposure and intestinal tumor frequency were scored and analyzed. Our analysis of tumorigenesis rate data clearly indicated a remarkable acceleration in tumorigenesis rate at the late time point in 28Si exposed mice, relative to gamma-rays and control groups. 5. Addition of acute neutron beam to our range of single beam GI tumorigenesis data: Male Apc1638N/+ mice (6-8 weeks) were whole-body exposed to sham-radiation, gamma-rays, 4He, 12C, 16O, 28Si, and 56Fe-ion at 10 cGy dose at NSRL, while neutron (10 cGy) exposure was conducted at Radiological Research Accelerator Facility (RARAF) (Columbia University). Mice were sacrificed at 150 d after radiation exposure and intestinal tumor frequency was scored

Task Progress:

and analyzed. The highest number of tumors was observed after 28Si followed by 56Fe, neutron, 16O, 12C, and 4He radiation. No statistical difference in tumorigenesis count was evident between neutron, 16O, and 12C at 10 cGy dose.

Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation (Project-2, Georgetown University): Proliferative gastrointestinal (GI) tissue is radiation-sensitive, and heavy-ion space radiation with its high-linear energy transfer (high-LET) and higher damaging potential than low-LET gamma-rays is predicted to compromise astronauts' GI function. However, much uncertainty remains in our understanding of how heavy ions affect coordinated epithelial cell migration and extrusion, which are essential for GI homeostasis. The overall goal of this project is to understand the long-term effects of space radiation on normal gastrointestinal (GI) cells including stem cells. Coordinated epithelial cell migration is key to maintaining functional integrity and preventing pathological processes in gastrointestinal (GI) tissue and is essential for astronauts' health and space mission success. Pointwise details of progress made for persistent effects of space radiation on GI-tissues are provided below:

1. We studied space radiation effects on intestinal stem cells (ISC) in male Lgr5 (Lgr5-EGFP-IRES-creERT2) mice in C57BL/6J background using 0.5 Gy of gamma-ray or 28Si ions. Lgr5 mice were exposed to 0.5 Gy of 28Si-ion and GFP+ Lgr5 cells were sorted from the isolated intestinal cells using flow-cytometry. Sorted cells were analyzed on flow cytometer for mitochondrial and cellular reactive oxygen species (ROS) using Mitosox or CellRox staining. Higher intracellular and mitochondrial ROS was observed in the 28Si exposed group, relative to gamma-ray and control GI stem cells.

2. We also demonstrated an increased level of beta-gal (senescence marker) positive intestinal stem cells in the 28Si exposed group, relative to gamma-ray. This suggests a higher number of stem cells were undergoing senescence after 28Si relative to gamma-ray or control group. This also correlated well with an increased number of gamma-H2AX foci in GI stem cells after 28Si compared to gamma ray or control group.

3. Emerging evidence indicates that senescent cells acquire a secretory phenotype (SASP) and production of inflammatory cytokines that lead to altered intestinal tissue homeostasis. A key pro-inflammatory secretory cytokine IL8 (SASP marker) was co-stained with p21 (senescent marker) and the data shows an increased level of IL8 expression in senescent cells (high p21 marker) indicating acquisition of the senescence-associated secretory phenotype (SASP) in a subset of senescent ISCs after 28Si exposure compared to gamma ray or control group.

4. A time-dependent analysis of persistent stress was conducted on samples collected at 2, 5, and 12-months post-exposure that indicated a progressive increase in DNA damage (gamma -H2AX), senescence (p16 and p21), and SASP (IL8) cells after 28Si exposure compared to gamma ray or control group.

5. Characterization of heavy ion-induced persistent oxidative modifications to DNA: We detected and quantified the spectrum of oxidative DNA base damage species using gas chromatography (GC)/mass spectrometry (MS) techniques two months after radiation exposure. Levels of damaged DNA base were significantly higher after 56Fe irradiation relative to sham-irradiated control as well as gamma-irradiated samples.

6. We also assessed the status of the repair pathways involved in repairing damaged DNA bases. Expression of BER (base excision repair) and NER (nucleotide excision repair) genes involved in the repair of identified DNA damage species was assessed in irradiated as well as in control samples. 56Fe exposure caused significant downregulation of many BER (Ogg1, Nth1, Neil1, Neil2, Neil3, Ape2, Xrcc1, and Lig3) and NER genes (Rbx1, ddb1, Ddb2, Xpc, Ercc8, Xpa, and Xpf). Additionally, immunoblot analysis of both BER (OGG1, NEIL1) and NER (DDB1, RBX1, CSB, NEIL1, and ERCC8) proteins showed significantly lower expression after 56Fe, relative to gamma radiation. We demonstrated a quantitatively higher accumulation of base damages that was associated with greater downregulation of BER and NER pathways after 56Fe, relative to gamma radiation.

Quantitatively and qualitatively compare effects of GCR-type irradiations on lung cancer initiation and progression in lung cancer susceptible mice. (Lead: Jerry W Shay) (Project-3, UTSW): The UTSW (University of Texas Southwestern) component of this project involves lung tissue from normal WT mice, and a mouse model of lung cancer (LA-1). The rationale for these models is that whole body exposure to space radiation will affect organ/tissues and cells in different ways. The lung only turns over cells with damage, while the colon turns over in humans every 7 days irrespective of damage. Thus, to model increased risks of more lethal cancers from space radiation exposure it is important to examine tissues with both a high rate of cellular turnover and tissues with a low rate of cell turnover. The inclusion of a lung model reflects the major solid tumor risk for astronauts traveling in space where the radiation-induced lung cancer incidence risk was estimated to be approximately 1% and 2.5% for male and female astronauts on a Mars mission, respectively, with upper 95% CIs extending well above 3% in both cases.

The overall hypothesis for this project is that space radiation induces molecular manifestations of a pro-inflammatory response as a function of radiation type, radiation doses, doses rates, LET value, and time that leads to increased risk of cancer initiation and progression. There remain uncertainties about cancer risks with single beam experiments and even simplified GCR simulations, but most experiments at the NSRL suggest late cancer effects are likely. It can be argued that we cannot predict what would happen in a mixed beam of fixed doses based on current data since most experiments were done in the acute setting with higher dose rates. While we recognize that there are issues of scaling from a mouse to human, and actually phenocopying the continuous low dose rates in the space environment, we have a large amount of legacy radiation carcinogenesis data and molecular baselines with mouse models using single beam experiments, a simplified 3 beam fast switching series of experiments, and now experiments with the 33-beam official full spectrum GCR simulation. We observed several hallmarks of cancer (even using a total dose of 30 cGy) in both our lung and colon cancer susceptible mouse models. Since our original NSCOR was focused on colon cancer, we had several mouse models of susceptibility to colon cancer and several papers have been published. However, the former space radiation element scientist recommended we add a lung cancer model for the GCR experiments in 2018 as part of our NSCOR.

We were the first team testing the NSRL-GCR 33 beam simulation with our lung cancer susceptible mouse model compared to WT mice starting in late 2018. Results from these studies were designed to evaluate the effects of both acute and chronic doses of the GCR simulator on radiation-induced health risks using a radiation-induced lung cancer in male and female wildtype and LA1 mutant (a lung cancer susceptible mouse) mice on a 129 background.

Each study included an acutely exposed group, a fractionated exposed group with consecutive doses of 20.8 mGy/day delivered over a four-week period, and sham controls. The simulator was baselined to 50 cGy for the first series of experiments. Acute exposures were delivered half-way through the fractionated exposure to support age-matching post irradiation During this initial experimental campaign approximately 300 mice received irradiations and ~200 mice

	treated as sham controls. The NSRL 18C (Fall 2018) run was a highly valuable experience in establishing concept of operations for future studies and providing a realistic determination of set-up and run times for future schedule planning. Cancer risk modeling (Columbia University) (Project-4, CU): A trip to Mars and back, or another lengthy space exploration mission, will result in exposure of astronauts to a multi-component mixture of ionizing radiations which damage cells in multiple ways. Such damage mechanisms can be roughly classified into two categories: (1) Targeted effects (TE), involving the consequences of direct traversals of cells by ionizing tracks leading to DNA double strand breaks and other lesions. (2) Non-targeted effects (NTE, or “bystander” effects) caused by release of signals from cells directly hit by the tracks and the effects of these signals on other cells. The presence of both types of effects generates complex dose response shapes for space radiations, particularly in the low dose region relevant for space missions. Here we used our mechanistically-motivated model formalism that includes both TE and NTE to analyze updated mouse tumorigenesis data from the current NSCOR. The updated data set now includes more detailed information on space-relevant doses of multiple radiation types. This new information and improved modeling approach allowed better estimation of dose response shapes and radiation effectiveness metrics in the space-relevant dose region. Specifically, relative biological effectiveness (RBE) and radiation effects ratio (RER) values at such doses of heavy ions and protons were estimated based on the most detailed data set version and improved methodology for describing variability in tumor numbers per mouse. These improvements provided updated and enhanced estimation of cancer risks in astronauts.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
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